Lipid Nanoparticle (LNP) Applications

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Abstract

Genome editing with therapies using CRISPR shows potential in the treatment of various genetic diseases. Lipid nanoparticles (LNPs) are a novel method of drug delivery, popularized recently through vaccines for COVID-19. Lipid nanoparticles traditionally consist of four different types of lipids: an ionizable cationic lipid, a neutral helper lipid, cholesterol, and a PEGylated lipid [1]. There are multiple advantages to using lipid nanoparticles in gene therapy, such as the non-immunogenicity, scalability, and lack of restrictions with packaging [2]. Currently, some applications of lipid nanoparticles are that they are able to deliver nucleases, base editors, transposases/recombinases, and prime editors [5]. The stability and structure can be analyzed for LNPs with different analytical methods, to encompass and support the theorized model of an LNP, as well as to optimize the methods for formulation and storage.

Key words: ionizable lipid, formulation, gene editing efficiency

1. Introduction

Recently publicized to the general population, lipid nanoparticles (LNPs) have shown potential for a variety of purposes. The mRNA vaccines developed by BioNTech/Pfizer and Moderna for COVID-19, which are composed of lipid nanoparticles encapsulating mRNA, present a novel solution for vaccines. These vaccines have approximately a 95% efficacy and were the first mRNA vaccines to receive authorization for emergency use by the FDA and conditional approval by EMA [1]. mRNA-LNP vaccines present multiple benefits compared to other types of vaccines. However, the lack of stability of mRNA vaccines is one of the greatest barriers that needed to be overcome during their development.

Encapsulation efficiency and degradation of mRNA and the lipid nanoparticles are some of the ways that LNPs can be analyzed to determine their efficacy.

2. Components of a Lipid Nanoparticle

With various mRNA as the cargo of lipid nanoparticles, four different types of lipids form the main structure of lipid nanoparticles. These lipids are affected by factors such as pH and composition of the lipids, and they can affect the structure, stability, and efficacy of LNPs.

2.1 Ionizable Cationic Lipid

Within LNPs, ionizable cationic lipids hold a crucial role in protecting RNAs and facilitating their transport. At an acidic pH, ionizable lipids are positively charged, to condense RNAs into LNPs. At physiological pH, they are neutral, to minimize the amount of toxicity. After cellular uptake, protonation of these lipids can occur in the acidic endosome, and cone-shaped ion pairs that are incompatible with a bilayer can be formed as a result of interaction with anionic endosomal phospholipids. There are currently five major types of ionizable lipids used for RNA delivery, which are unsaturated, multi-tail, polymeric, biodegradable, and branched-tail, and they can be categorized by their structures [3].

2.2 Neutral Helper Lipid

Neutral helper lipids generally provide support to LNPs through stabilizing effects, blood compatibility, and to allow for enhanced efficiency of oligonucleotide delivery. A couple examples of common helper lipids include 1,2-Dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE), which is depicted by Figure 1, and Phosphatidylcholine (PC) [4]. DOPE has been used as a helper lipid to facilitate endosomal escape, however, LNPs with DOPE have lower colloidal stability. PCs can be used within LNP formulations to increase stability and decrease toxicity in plasmid delivery [4].

2.3 Cholesterol

Furthermore, another lipid that is commonly included in LNP formulation is cholesterol. Similarly to helper lipids, it is commonly used for stability purposes of lipid bilayers and for enhancement of cationic lipid activity. Its inclusion in LNP formulations for gene delivery *in vivo* has been found to provide better support than DOPE, and its presence with PC results in stable lipid bilayers commonly used in oligonucleotide LNP formulations [4].

2.4 PEGylated Lipid

PEGylation is the incorporation of a PEGylating lipid into liposomes and LNPs. It can enhance stability and resistance to clearance *in vivo*. The most common PEGylating lipid used in formulations of liposomes and LNPs is mPEG-DSPE. A disadvantage of PEGylation is that it can inhibit cellular internalization and intracellular release of oligonucleotides. For this reason, when formulating LNPs, using a low mole percentage of mPEG-DSPE can help mitigate this issue [4].

3. Current Analysis of Lipid Nanoparticles

With lipid nanoparticles (LNPs) being difficult to analyze due to the small size, the exact structure is not known and only theorized. The various lipids within LNPs can also contribute to the formulation of LNPs, their stability and their efficiency. Stability can also be analyzed over time of the LNPs and the mRNA, and different storage and formulation conditions can contribute to this.

3.1 Structure of Lipid Nanoparticles

The structure of LNPs used for mRNA and siRNA delivery differ due to differences in size and structure of the encapsulated cargo. Figure 3 depicts three models for mRNA-LNPs: a multilamellar vesicle model (left of Figure 3) or a core-shell model (middle and right of Figure 3). Most researchers believe that mRNA-LNPs have a core-shell structure, with mRNA located in the amorphous, isotropic core surrounded by water cylinders and cationic lipids, while siRNA is more towards the surface [1].

3.2 Stability of Lipid Nanoparticles

LNPs and their integrity can be tested for stability through various methods, such as gel electrophoresis, fluorescence correlation spectroscopy (FCS), and HPLC. Gel electrophoresis is a technique that allows for the size and integrity of the mRNA to be tested. FCS allows for changes in the size of mRNA to be monitored when there are substantial changes in molecular weight, and it requires a fluorescent label. HPLC techniques, such as RP-HPLC, SE-HPLC, IP-HPLC, and IEX-HPLC have also been developed [1]. These various techniques have shown that chemical and physical instability can occur in LNPs. Chemical instability includes the degradation of lipids in the LNPs due to hydrolysis and oxidation. Oxidation can occur either in unsaturated fatty acid moieties and with cholesterol or oxidation of mRNA, due to impurities. Hydrolysis can also occur in the carboxylic ester bonds in lipids such as Distearoylphosphatidylcholine (DSPC) and ionizable cationic lipids, due to effects of temperature and pH [1].

3.3 mRNA Stability

The stability of mRNA and LNPs has been explored, but it is inconclusive whether mRNA or LNPs are more unstable. Research has shown that LNPs containing stable siRNA have longer shelf lives than mRNA-LNP systems, expressing that mRNA may be limiting the current stability. There is limited research on the long-term stability of mRNA and LNPs in mRNA-LNP formulations. Studies suggest that naked mRNA needs to be frozen or dried to remain stable for extended periods of time and that mRNA can only be stored in aqueous solutions for a few days at 4°C when protected from degradation. The stability of mRNA in LNPs is also uncertain, as mRNA in LNPs is inherently an aqueous environment and may be susceptible to hydrolysis. Without further research, it can only be concluded that mRNA instability has an effect on conditions for storage and that further studies could allow for optimization of formulations [1].

3.4 Storage of Lipid Nanoparticles

In addition to the stability of LNPs due to the formulation process through composition of the lipids, storage of the formulated LNPs can also affect the stability and structure. Long-term storage of LNPs can limit the efficacy, due to increasing instability over time. Some methods used to increase stability with storage are low temperatures and nanoparticle lyophilization. A loss of efficacy has been seen with storage at room temperature, in a study by Ball, et. Al., from Carnegie Mellon University [6], which could possibly have been caused by hydrolysis of the ester bond within the lipid. Furthermore, it was found that alteration of the pH of the PBS solution that the LNPs were stored in did not affect the properties or potency of the LNPs at any temperature. Moreover, freeze-thaw cycles are a viable option to test for LNP storage, and this same study saw that freezing LNPs with sugar can decrease an increase in size and polydispersity index (PDI) of the LNPs. Lyophilized LNPs were also tested and determined to have lost efficacy after reconstitution with DI water. Thus, storage of LNPs can be further explored to more clearly determine the optimal method for maximizing efficacy and gene expression with LNPs used for drug delivery.

4. Conclusions, challenges, future directions

This review discusses the recently popularized genetic disease treatment, lipid nanoparticles, what they are composed of, and the various factors that define their formulation. LNPs primarily structurally consist of four lipids that contribute to their overall formulation and efficacy: ionizable cationic lipids, neutral helper lipids, cholesterol, and PEGylated lipids. The ionizable cationic lipid within LNPs largely contributes by protecting RNAs and allowing for their transport. Both the neutral helper lipid and cholesterol contribute by providing stability to lipid bilayers and enhancing the activity of cationic lipids. Furthermore, PEGylated lipids in LNPs increase the stability and transport of LNPs *in vivo*. Additionally, LNPs can be analyzed by various methods to determine how the stability of mRNA compares to the LNPs, through storage and other formulation conditions. Further research of LNPs for gene delivery could lead to new therapeutic opportunities for treating genetic diseases, improvements of their stability, and improvements of their structure for more efficient and effective delivery of genetic therapies.

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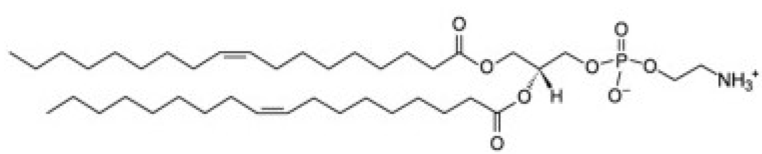
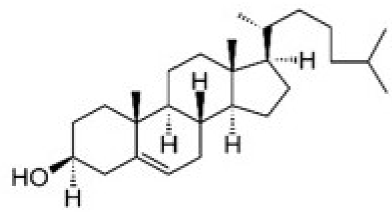
Figure 1



DOPE

Figure 1. Molecular structure of the helper lipid 1,2-Dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE) [4]

Figure 2



CHOL

Figure 2. Molecular structure of the helper lipid cholesterol [4]

Figure 3

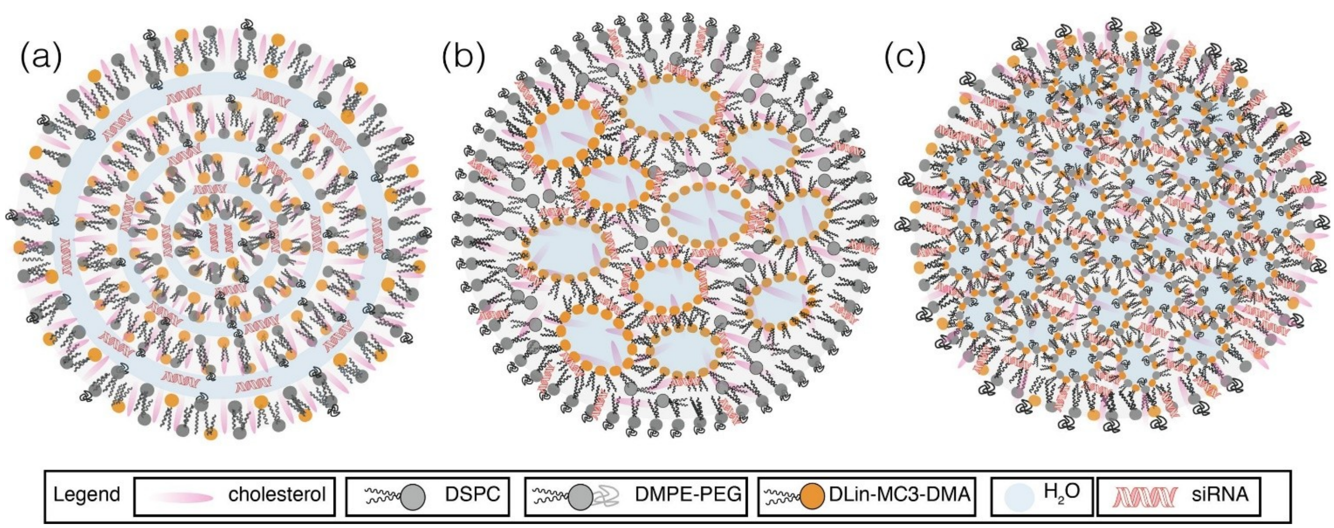


Figure 3. Theorized structures of LNPs [1]

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